



Southwest WI Technical College

10-152-118 HTML, CSS & JavaScript

Course Design

Course Information

Alternate Title	Web Foundations for AI-Augmented Engineers
Description	Learners develop foundational web development skills using HTML, CSS, and JavaScript to create structured, styled, and interactive web pages. The course emphasizes building multi-page websites, applying design principles for usability and readability, and implementing basic interactivity through JavaScript. Learners practice debugging and refining web applications using progressive techniques and are introduced to AI-assisted tools to support development. By the end of the course, students are able to design, build, and explain functional web solutions.
Career Cluster	Information Technology
Instructional Level	A.A.S. - Associate in Applied Science
Total Credits	3
Total Hours	72

Types of Instruction

Instruction Type	Credits/Hours
(A) Lecture	36
(B) Lab	36

Textbooks

Responsive Web Design with HTML5 and CSS: Build future-proof responsive websites using the latest

JavaScript and jQuery: Interactive Front-End Web Development - 1st Edition - Jon Duckett

HTML 5: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

CSS - Cascading Style Sheets: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

Course Competencies

1. Structure web content using HTML

Assessment Strategies

- 1.1. Project assignments and quiz

Criteria

- 1.1. Create structured web pages using semantic HTML elements
- 1.2. Organize content logically using headings, paragraphs, lists, and links
- 1.3. Build multi-page website structures
- 1.4. Apply consistent page organization
- 1.5. Ensure structure supports styling and interactivity

Learning Objectives

- 1.a. Define HTML structure and purpose
- 1.b. Identify common HTML elements
- 1.c. Explain semantic markup
- 1.d. Describe how HTML structure supports styling and interaction

2. Design and style web pages using CSS to create readable, usable, and responsive layouts

Assessment Strategies

- 2.1. Project assignments and quiz

Criteria

- 2.1. Apply CSS to control layout, spacing, typography, and color
- 2.2. Use selectors, classes, and IDs effectively
- 2.3. Create visually consistent and readable designs
- 2.4. Apply basic responsive design techniques
- 2.5. Ensure styling supports usability and interaction

Learning Objectives

- 2.a. Define CSS styling principles
- 2.b. Explain layout concepts (box model, positioning)
- 2.c. Identify responsive design basics
- 2.d. Explain how styling impacts usability and user experience

3. Implement interactive behavior using JavaScript to respond to user input and manipulate web page content

Assessment Strategies

- 3.1. Project assignments and quiz

Criteria

- 3.1. Write basic JavaScript using variables, conditionals, and functions
- 3.2. Implement interactive behaviors in web pages
- 3.3. Respond to user input and events
- 3.4. Connect program logic to visible outcomes

3.5. Manipulate HTML elements and styles dynamically

Learning Objectives

- 3.a. Define JavaScript as a programming language
- 3.b. Explain variables, functions, and control flow
- 3.c. Describe how JavaScript enables interactivity
- 3.d. Explain how JavaScript interacts with HTML and CSS

4. Debug and evaluate web applications using progressive techniques across HTML, CSS, and JavaScript

Assessment Strategies

- 4.1. Project assignments and quiz

Criteria

- 4.1. Identify and correct layout and styling issues
- 4.2. Use console logging to trace JavaScript behavior
- 4.3. Utilize browser tools to inspect and troubleshoot web pages
- 4.4. Apply structured debugging approaches to resolve errors
- 4.5. Diagnose issues across HTML, CSS, and JavaScript interactions

Learning Objectives

- 4.a. Define debugging and troubleshooting
- 4.b. Identify common HTML, CSS, and JavaScript issues
- 4.c. Explain debugging strategies
- 4.d. Describe how issues can originate across multiple layers

5. Develop web solutions using structured problem-solving, appropriate tools, and iterative improvement

Assessment Strategies

- 5.1. Project assignments and quiz

Criteria

- 5.1. Plan and build web pages based on defined requirements
- 5.2. Apply structured problem-solving to design and implementation
- 5.3. Use AI tools to assist in development when appropriate
- 5.4. Evaluate and refine solutions for usability and clarity
- 5.5. Iteratively improve solutions based on testing and feedback

Learning Objectives

- 5.a. Describe structured development approaches
- 5.b. Identify when and how to use AI tools
- 5.c. Explain iterative improvement of web solutions
- 5.d. Explain how planning, building, and refining form a development cycle